Heaven's Light is Our Guide

Rajshahi University of Engineering and Technology

Department of Electrical and Computer Engineering

Course No. : ECE 2100

Course Title : Software Development Project- I

Project Title: Game

Project Name: Circumvent Bars

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Introduction :

This project is suggest to us under the academic course ECE 2100 in Second year odd semester course. This project is supervised by Sagor Chandro Bakchy, Assistant Professor , Department of Electrical and Computer Engineering, RUET and Md. Mahabubur Rahman, Assistant Professor , Department of Electrical and Computer Engineering, RUET.

In this project I tried to make a game mainly for entertaining purpose using 'Unity-Engine' and C sharp programming language.

Using Unity-Engine the designing and the graphics work become easier.

This is a gaming project. There are four levels in the game. Each level comes with new speciality. This a cross platform game . So , it can be played both mobile phone and pc.

Software Tools :

In this project I've used Unity Engine version 2021.1.0b1.1725 . For programming I've used C# language which required Visual Studio version 16.10.0 . For transferring the game into android version I've used Java Development kit (JDK) version 12.0 and JRE version 1.8.0_111.

Hardware description:

This project is developed in the environment of Wnidows10 operating system. This is a 64-bit operating system. The hardware in this system is follows

1.Processor: In this system there is a Intel core™ i3-5005U processor which has a average processing speed 2.00 GHz.

2.RAM: The physical memory(RAM) installed in this system is total 12.00GB DDR3 SDRAM.

3. Graphics property: In the system, the installed Graphics adapter is Intel(R) HD Graphics 5500. Which provides a total Available Graphics Memory 6229 MB and Dedicated Video Memory of 128MB. This Graphics adapter also provides Shared System memory 6101MB.

4. HDD and Device: This system runs in a Laptop computer of model "HP NOTEBOOK F.14" made by the Hewlett Packard Company Pvt. Ltd. The available HDD in the system is 464 GB.

Purpose of the Project:

This project is mainly created for entertaining purpose. Gaming is always fun. There are also other benefits of playing games:

- Studies show that kids who play video games may get a small boost to their reading skills.
- Video games can improve manual dexterity.
- Video games can increase your brain's grey matter.
- Games can teach you to be a better problem solver.
- Video games can have mental health benefits.

The details of the game and how should a player should play it is given below.

Details of the Game:

The name of the game is 'Circumvent Bars'. Avoiding obstacles and reach the goal is the theme of the game. There are four levels in the game. The detailed features of the game is given below.

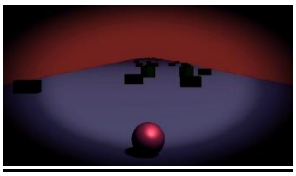

Figure01: Logo of the game

This is the icon or the logo of the game. The user will see it as app icon to desktop or at phone.

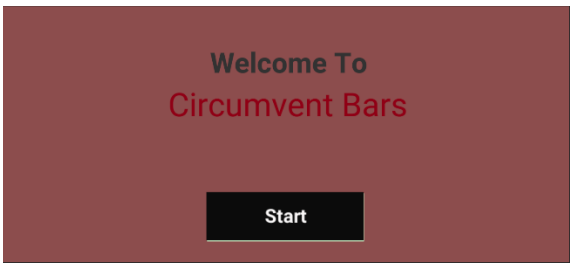

Figure02:Starting page of the game.

This is the starting page of the game. By clicking the start button the main menu page will appear . Music and animations are added in this segment.

Figure03:Start button will take the player to menu

Menu Page:

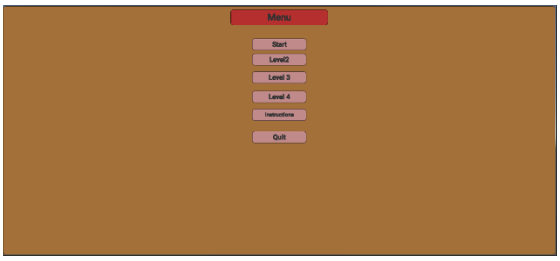

Figure04: Menu Page

There are six buttons in the menu. The each button will take the player to a new page

- Start Button: The start button take the player to level 1

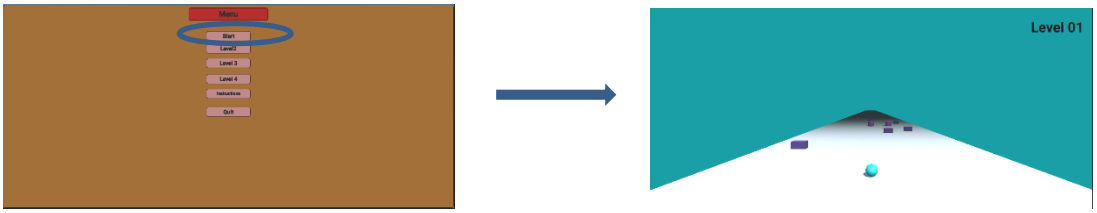

Figure05:Working of start button

- Leve2 Button: The Level2 button take the player directly to level two of the game.

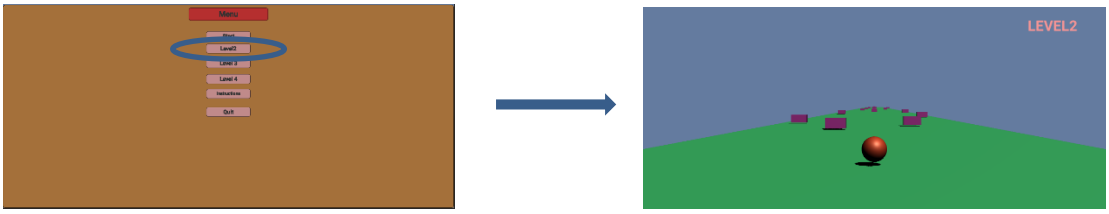

Figure06:Working of 'level2' button

- Level3 Button: The Level3 button take the player directly to level three of the game.

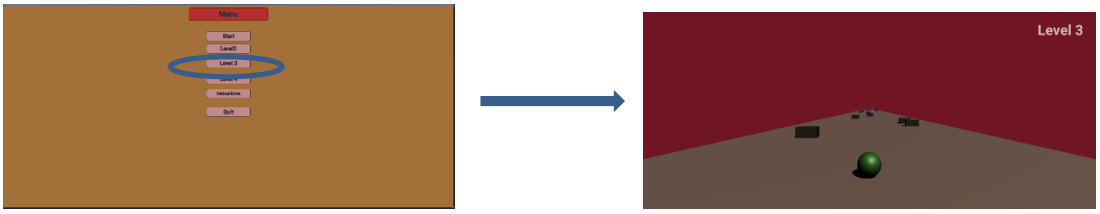

Figure07:Working of 'level3' button

- Level4 Button: The Level4 button take the player directly to level four of the game.

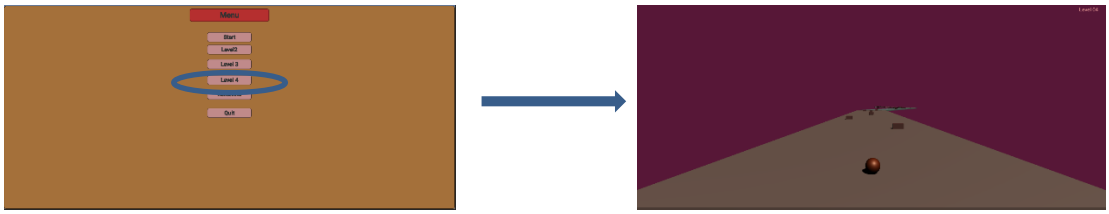

Figure08:Working of 'level4' button

- Instruction Button: The instruction button take the player to the instructions page where a player can know how to play the game.

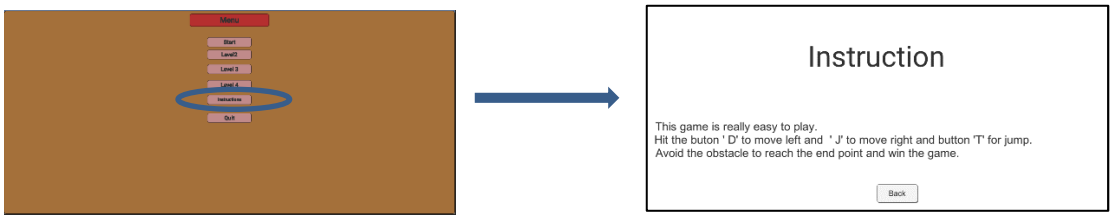

Figure09:Working of 'Instructions' button

There is a 'Back' button in the instruction page. The button will take the player to the main menu page.

- Quit Button: The Quit button of the menu quits the game.

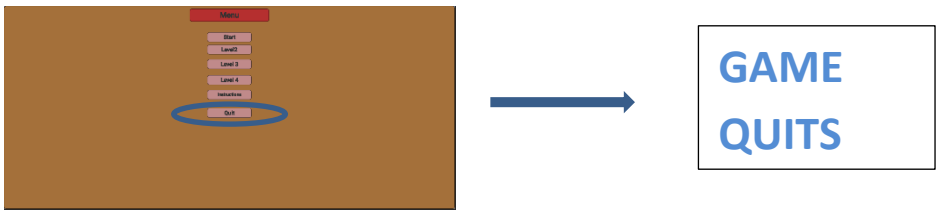

Figure10:Working of 'Quit' button

Gaming Details:

- There are obstacles and a ball which should avoid the obstacles to reach the end point of the level.

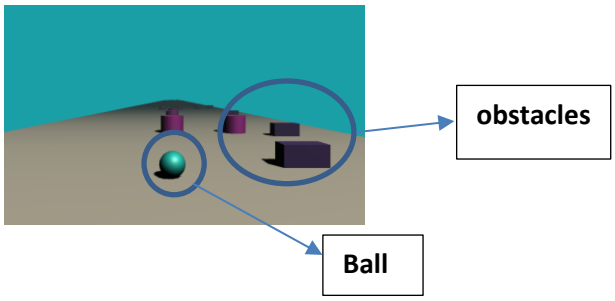

Figure11:Ball and Obstacle

The player shoud use button ‘D’ or button ‘<’ to move the ball left, button ‘J’ or ‘>’ to move the ball right and button ‘T’ or ‘^’ to jump the ball. By using these a player can control the ball.

- If the ball fall from the path the game will restart from the existing level.

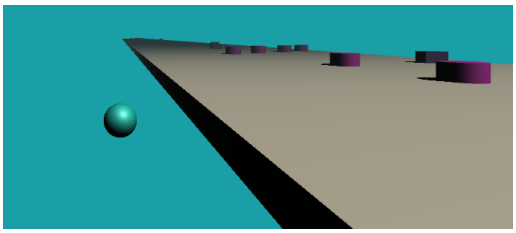

Figure12: Ball falling from the path

- The score is being showed at the top right corner.

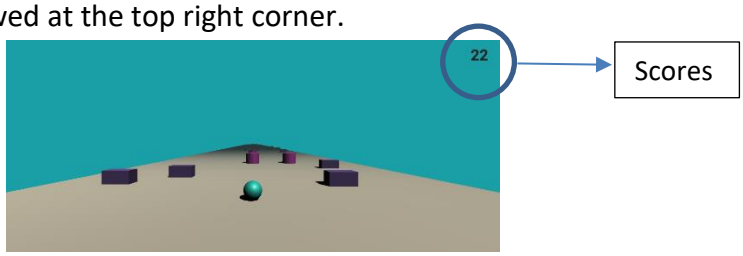

Figure13: Showing Score

- While the player reaches the end point of the game a page saying “Congratulations Level Complete” will appear. Animations are added in this segment.

Figure14:Level completing page

After the completing level page the next level will reload automatically .

- If the player collides with the obstacle , the game will over and a page saying game over will appear. There are two buttons ‘Back’ and ‘Quit’ in this segment.The ‘Back’ button takes the player to the main menu and the ‘Quit’ button quits the application.

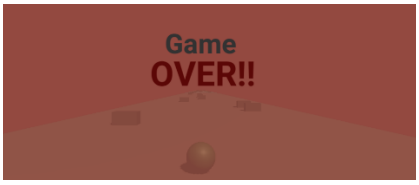

Figure15:Game Over

- If a player can complete all the four levels a page saying ‘Game Won’ will appear at the last. Animations and music are added in this page.

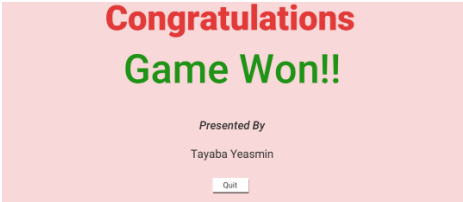

Figure16:Winning Game

The ‘Quit’ button in this page will quit the game

Versions:

The game is a cross plartform game. So, it can be played both in mobile phone and pc. In pc the game can be played at windows,Linux and Mac operating system. And in mobile phone the game can be played in android , ios or any other platform.

Differences Between Mobile Phone and PC Version: There is approximate no difference in visualising the Mobile Phone and the PC version. The main difference is in pc version the ball can be controlled using the key ‘D’,’J’ and ‘T’. But in the Mobile phone version there will three buttons in the screen ‘<’,’>’, and ‘ ^ ‘ which will control the ball.

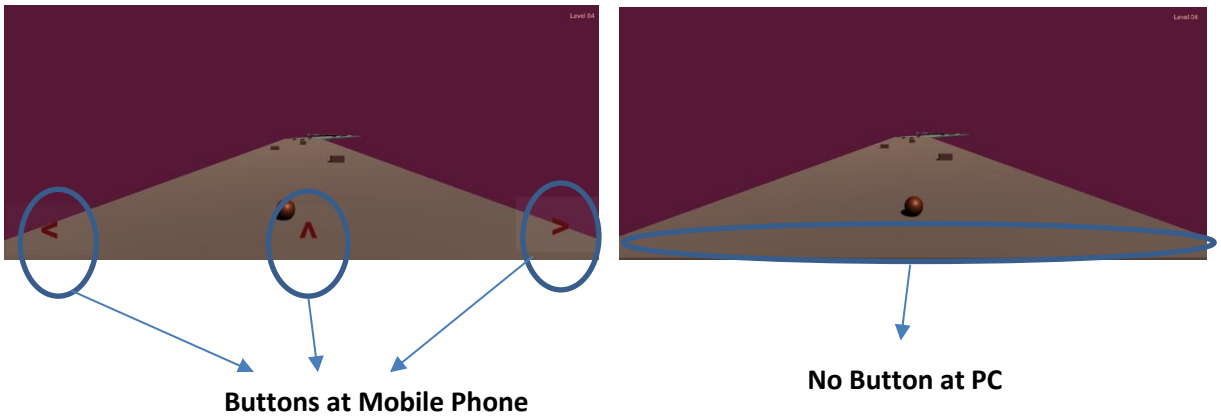

Figure17: Difference between mobile phone and pc

Conclusion and Discussion:

Making a game is always challenging. To build this project a few problems have been faced. Although at the end the game run successfully. This is a simple game and the game is really easy to play. One can play it by only going through its instructions. This game is really fun to play and not so much time consuming like other video games. This projct is a simple souece of having entertainment.